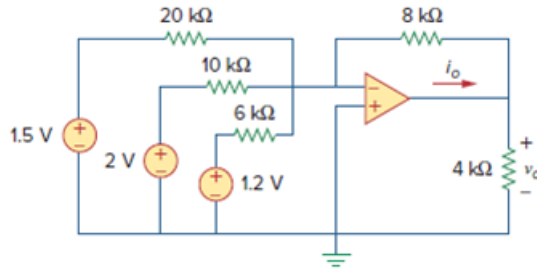


Problem

Find v_o and i_o in the op amp circuit shown in Fig. 5.23.

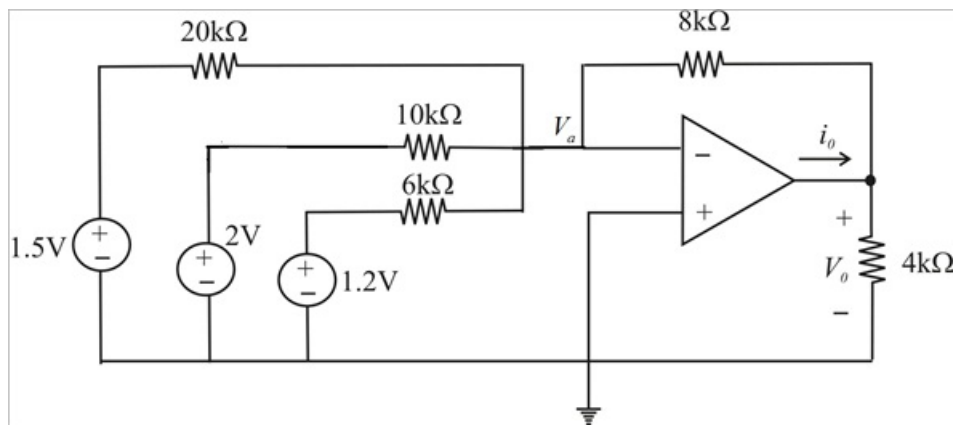
Figure 5.23:



Step-by-step solution

Step 1 of 3

Given OP amp circuit



Step 2 of 3

The above circuit is inverting summing amplifier thus the output V_o is

$$V_o = - \left[\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right]$$

From the circuit we have the V_o

$$V_o = - \left(\frac{8k}{20k} \times 1.5 + \frac{8k}{10k} \times 2 + \frac{8k}{6k} \times 1.2 \right)$$

$$V_o = - (0.4 \times 1.5 + 0.8 \times 2 + 1.33 \times 1.2)$$

$$V_o = - (0.6 + 1.6 + 1.596)$$

$$\boxed{V_o = -3.796 \text{ V}}$$

Step 3 of 3

By applying KCL at node (1)

$$\frac{V_0 - V_a}{8k} + \frac{V_0}{4k} - i_0 = 0$$

We have $V_a = V_b = 0$

$$\frac{V_0 - 0}{8k} + \frac{V_0}{4k} - i_0 = 0$$

$$\frac{V_0}{8k} + \frac{V_0}{4k} = i_0$$

$$0.125 \times 10^{-3} V_0 + 0.25 \times 10^{-3} V_0 = i_0$$

$$0.375 \times 10^{-3} V_0 = i_0$$

$$0.375 \times 10^{-3} \times (-3.796) = i_0$$

$$\boxed{i_0 = -1.4235 \text{mA}}$$